How much does local inference cost?

Qwen2.5 on edge hardware | Mac phase; Android pending

- Two Qwen2.5 Instruct sizes: 0.5B and 1.5B
- Weight variants: Q4_K_M and Q8_0
- Apple M1 CPU baseline; Android replication later

A fast decoder can still feel slow

LOAD	PREFILL	DECODE	DELIVERY
process start to model ready	evaluate the formatted input	generate at the populated context	includes request and stream delivery

We narrowed the pair after feasibility research

- Original proposal: TinyLlama 1.1B + Qwen2.5 3B
- Adopted: Qwen2.5 0.5B + 1.5B Instruct
- Same family improves control; architecture still changes
- Phone RAM estimate: 6-8 GB, not yet verified

The Mac matrix has 24 configurations

- 2 model sizes × 2 weight variants
- 3 input lengths: 256 / 1,024 / 2,048 tokens
- 2 output budgets: 128 / 512 tokens
- Mac presentation pilot: 1 observation/config (n=1)

Equal labels need exact token workloads

- Count tokens after all formatting
- Save exact input arrays and actual output counts
- Capacity: 3,072 positions; largest occupied: 2,560
- No prompt reuse; forced output caps disclosed

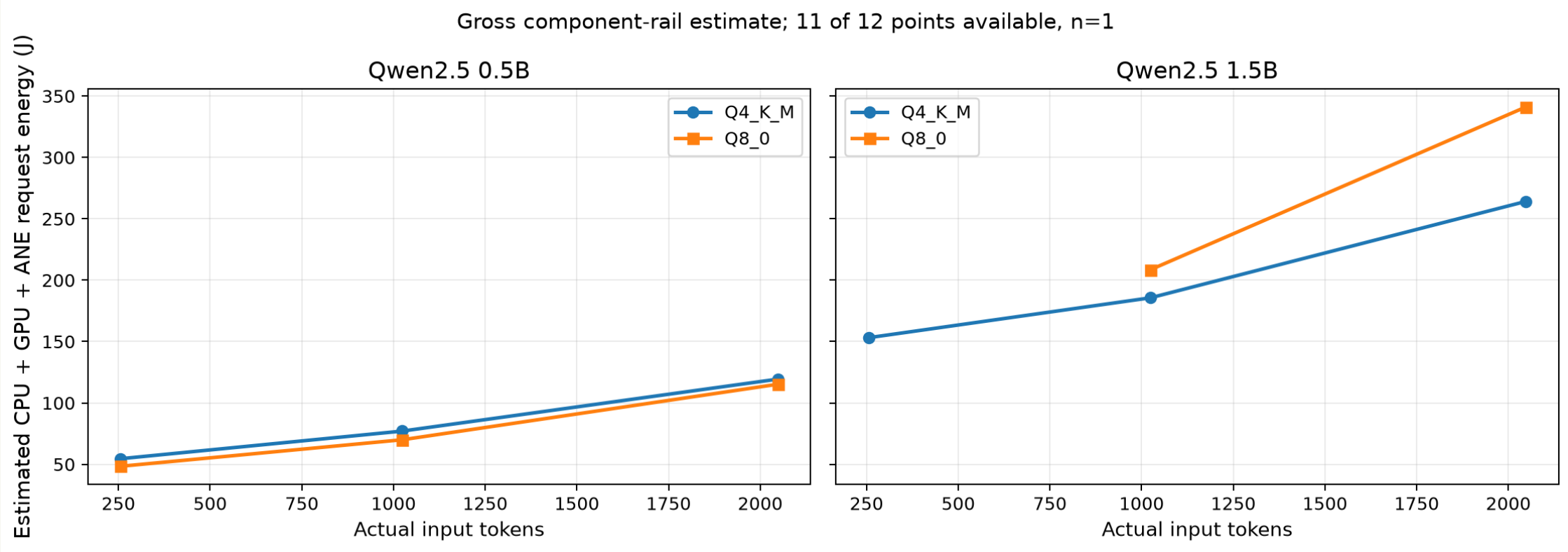
Pin the experiment, then prove the backend

- Model revision + file SHA-256
- llama.cpp source commit + build options + binary hash
- CPU, 4 threads, one request; zero GPU layers
- Raw logs → run records → analysis → artifacts

Thermal data are exploratory in this presentation run

- One bounded temperature observation per model block
- Reference window: 40.85–50.25 °C; not stabilized
- Continuously record actual temperature and nominal pressure
- Strict 60 s / ≤1 °C gate stays in the full protocol

Power boundaries decide which claims are valid



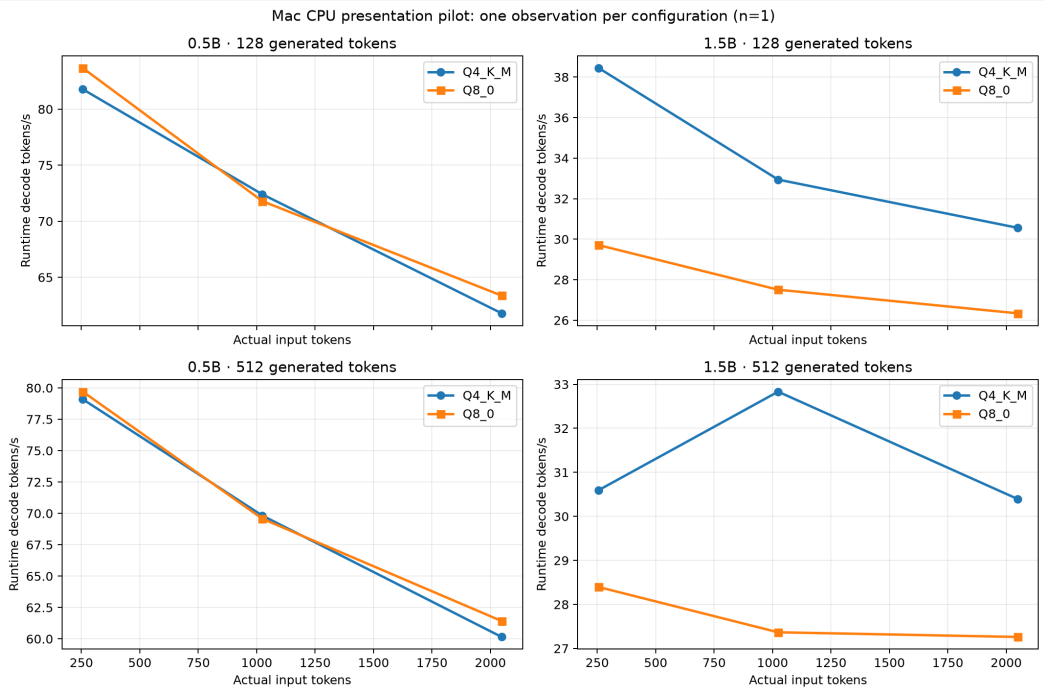
512-token requests: 11/12 have bracketed estimates (48.4-341.1 J), n=1; gross CPU+GPU+ANE only.

No whole-device or decode J/token estimate; Android USB boundary remains pending

Results begin with coverage

- Mac: 24/24 valid observations (one per configuration, n=1); Android: 0/24 pending.
- Warm-up, pilot, exploratory and failure records stay separate
- Android: pending teammate execution

Weight quantization: compare within each checkpoint

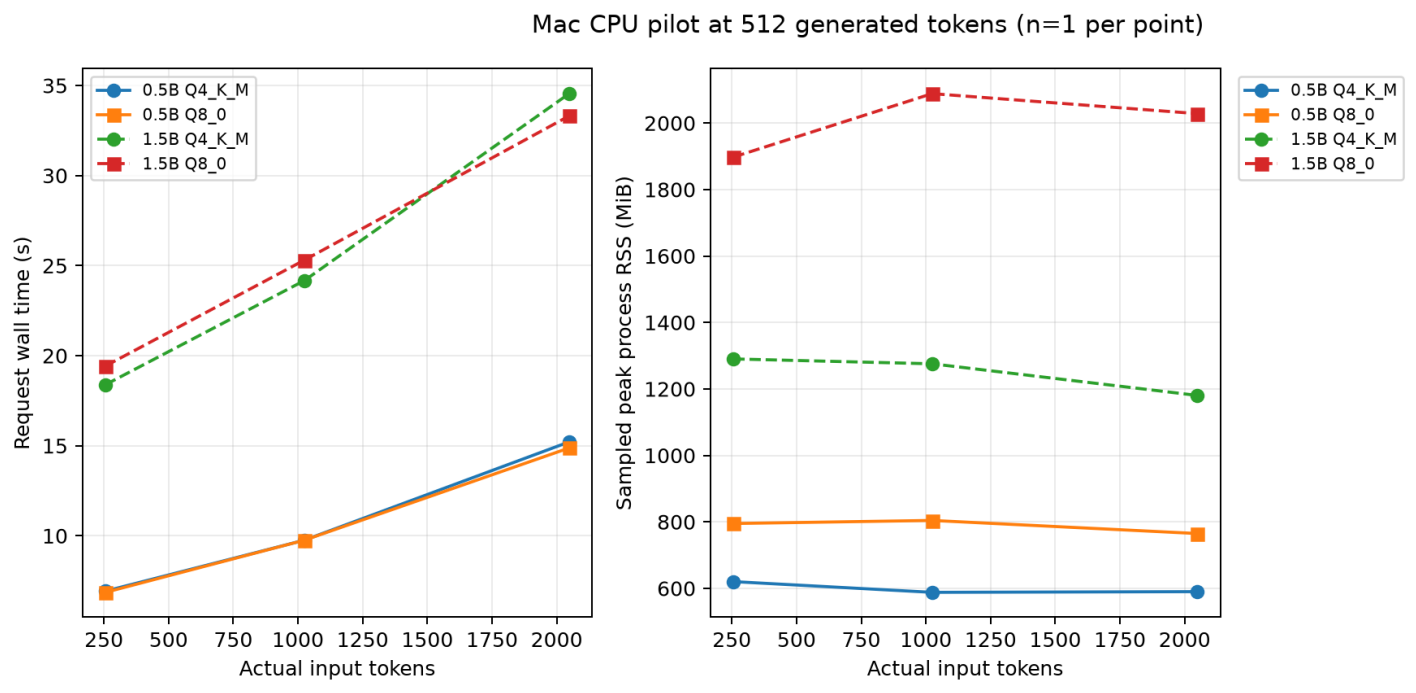


Mac CPU decode observations span 26.34-83.66 tokens/s across different model, quantization, input, and output settings. Every point is n=1; compare only matched configurations.

Q4 and Q8 are compared within each checkpoint

One observation per point; no SD or error bars

Context changes both work and interpretation



Prefill rates were 110.5-597.0 tokens/s across 256-2,048 input tokens. Capacity was 3,072 positions; input length is not allocated cache.

Prefill and decode remain separate

Allocated cache capacity stays fixed at 3,072

Focused tests answer different questions

- KV-cache comparison: not run in this presentation sweep
- WikiText-2 perplexity: not run
- Operator profiling: not run
- No sub-study results are implied by the main matrix

What the evidence can support

- Mac main matrix: one observation/configuration
- No SD, significance, or repeatability claim
- Android evidence is still pending
- Conclusions are limited to tested CPU conditions

Demo: trace one real request to its evidence

- Play the verified video (< 120 s)
- Show exact configuration and real saved response
- Open raw timing + token-count record
- Locate the same run in the analysis

Android handoff preserves the experiment

- Record phone identity, power state, sensors and units
- Same source + model hashes; CPU baseline verified
- Small smoke → largest pilot → cooled randomized matrix
- Return raw evidence, failures and configuration manifest

The contribution is an auditable testbed

- Every performance claim points to a measured run
- Phase timing, memory and power keep their boundaries
- Mac evidence now; two-device conclusions after replication
- github.com/addiee13/project_effAI

References and questions

- Qwen2.5 Technical Report · arXiv:2412.15115
- Official Qwen 0.5B / 1.5B GGUF repositories
- Pinned llama.cpp + llama-bench documentation
- WikiText · arXiv:1609.07843